5th International Conference on Advanced Security of Software and Systems ASSS 2026 Conference Handbook

About the Conference

ASSS 2026 focuses on software and systems security, providing a forum for researchers from academia and industry to exchange emerging technologies, research findings, and practical experience. Topics include artificial intelligence security and AI applications in security, software vulnerabilities and defense, malware analysis, intrusion detection, program analysis, blockchain security, formal methods, and application security.

Dates Venue and Attendance

Dates: 14–15 November 2026

Venue: Linyi Chengfa Jinjiang International Hotel, 101 Lanting Road, Hedong District, Linyi, Shandong, China.

Format and attendance: Hybrid on 14 November and online on 15 November; 11 contributed presentations and approximately 30 participants.

Organization

Organizers: Linyi University; SPTAGE Lab (UK); Springer.

General Chairs: Weizhi Meng (Lancaster University, UK); Qingni Shen (Peking University); Wenjuan Li (The Education University of Hong Kong).

Program Chairs: Jiuru Wang (Linyi University); Chunhua Su (University of Aizu, Japan); Guangdong Bai (City University of Hong Kong).

Publicity Chairs: Jiale Zhang (Yangzhou University); Wei-Yang Chiu (Technical University of Denmark); Jingxue Chen (Nankai University).

Publication Chair: Hongyuan Cheng (Linyi University). Local Organizing Chair: Jingcheng Song (Linyi University).

Conference website: https://asss.sptagelab.org/2026/

Registration

Friday 13 November 2026

14:00–18:00 Participant registration

Location: Linyi Chengfa Jinjiang International Hotel.

Conference Program

Saturday 14 November 2026

Morning 08:00–12:00 | All times are China Standard Time (UTC+8)

Time	Session	Talk Title or Activity	Presenter	Chair	Min.
08:00–08:10	Opening	Conference introduction by the host	Jiuru Wang	Jiuru Wang	10
08:10–08:40	Opening	Leadership addresses		Jiuru Wang	30
08:40–08:50	Opening	Group photo	All participants	Jiuru Wang	10
08:50–09:30	Keynote 1	When AI Becomes the Attack Surface: Security and Privacy Challenges of Foundation Models	Chunhua Su University of Aizu	Hongtao Li	40
09:30–10:10	Keynote 2	Dynamic Symmetric Searchable Encryption with Conjunctive Queries and Malicious Server Traceability	Cong Zuo Beijing Institute of Technology	Hongtao Li	40
10:10–10:40	Coffee Break	Coffee break and networking	—		30
10:40–11:20	Keynote 3	A Safety Harness for Embodied Intelligence	Weizheng Wang Guangzhou University	Jiuru Wang	40
11:20–12:00	Keynote 4	Compression Techniques for Large Multimodal Models	Guolei Sun Nankai University	Jiuru Wang	40
12:00–14:00	Lunch Break	Lunch and rest	—		120

Conference Program

Saturday 14 November 2026

Afternoon 14:00–16:50 | China Standard Time (UTC+8)

Time	Session	Talk Title or Activity	Presenter	Chair	Min.
14:00–14:20	Paper 1	FssTQU: an FSS-based (τ,n)-threshold secure database query and update framework	Mingjun Sun	Chunhua Su	20
14:20–14:40	Paper 2	The Observable Attacker: Exact Abstraction for Discovery-Budget Security	Wenxin Fang	Chunhua Su	20
14:40–15:00	Paper 3	Fine-Grained Data Query and Access Control Scheme for MaaS	Yanlin Du	Jingcheng Song	20
15:00–15:30	Coffee Break	Coffee break and networking	—		30
15:30–15:50	Paper 4	A Verifiable Secure Federated Learning Framework for Energy Theft Detection	Zhixin Zeng	Jingcheng Song	20
15:50–16:10	Paper 5	DualDiff: Differentially Private Image Release via Latent Diffusion Models	Lele Yu	Jingcheng Song	20
16:10–16:30	Paper 6	Multi-Diff-MESA: Gradient-Stability-Guided Tensor Freezing for Efficient Horizontal Federated Learning	Chenrui Gu	Jingcheng Song	20
16:30–16:50	Paper 7	Detecting the anomaly traffic by using the improved double deep Q-network in IoT	Guihua Liu	Jingcheng Song	20

Conference Program

Sunday 15 November 2026

Online session 09:00–10:20 | China Standard Time (UTC+8)

Time	Session	Talk Title or Activity	Presenter	Chair	Min.
09:00–09:20	Paper 8	Empirical Study on AI Hallucinations Based on Knowledge Graph	Wenjun Wu	Jingcheng Song	20
09:20–09:40	Paper 9	Falcon-Gated Kyber Decapsulation for Constrained Receivers Side-Channel Exposure Control and Protected Decryption	Bukhtawar Umbreen	Jingcheng Song	20
09:40–10:00	Paper 10	An intrusion Detection Scheme Based on Multiscale Spectral Residuals and Spatial-Temporal Parallel Learning for In-Vehicle Networks	Miaomiao Liu	Jingcheng Song	20
10:00–10:20	Paper 11	A Comparative Survey of JavaScript Obfuscation and Minification on Surface and Darknet Websites	Noriaki Yoshiura	Jingcheng Song	20

Online Participation

Platform: Tencent Meeting | Meeting topic: Academic Conference

Meeting ID: 754-230-065

Meeting reservation window: 15 November 2026, 09:00–17:00 (UTC+8)

Join link: https://meeting.tencent.com/dm/fudlyorVFHFO

Dial-in number (Mainland China): +86 (0)755 36550000

Keynote Speakers and Talks

Chunhua Su

Senior Associate Professor, University of Aizu, Japan

Saturday 14 November 2026 | 08:50–09:30 (UTC+8)

When AI Becomes the Attack Surface: Security and Privacy Challenges of Foundation Models

Abstract

Foundation models, particularly large language models, are becoming deeply embedded in software systems, the Internet of Things, and critical information infrastructure, evolving from tools to be protected into a new attack surface. Unlike traditional software, their behavior is jointly determined by data and training processes. Their security boundaries are unclear and their decision-making processes are difficult to verify, creating new security and privacy risks. This talk first reviews major threats throughout the foundation model lifecycle: data poisoning and backdoor insertion during training; prompt injection, jailbreaks, and adversarial examples during inference; and privacy leakage through model inversion, membership inference, and training data extraction. It also discusses system-level risks arising when foundation model agents invoke external tools. Drawing on the speaker's research in federated learning, secure multiparty computation, and privacy-preserving inference, the talk then explains how cryptographic techniques and distributed collaborative learning can build trustworthy AI systems that resist poisoning and protect privacy. It also covers guardrails and runtime detection mechanisms for foundation models. Finally, it explores open questions and future research directions in verifiable AI and the integration of privacy-preserving computing with foundation models.

Biography

Chunhua Su is a Senior Associate Professor at the University of Aizu, Japan. He received the B.S. degree from Beijing Electronic and Science Institute in 2003, and the M.S. and Ph.D. degrees in Computer Science from the Faculty of Engineering, Kyushu University, Japan, in 2006 and 2009, respectively. From 2009 to 2011, he was a Postdoctoral Fellow at Singapore Management University, and from 2011 to 2013, he worked as a Research Scientist in the Cryptography & Security Department of the Institute for Infocomm Research, Singapore. Between 2013 and 2016, he was an Assistant Professor at the School of Information Science, Japan Advanced Institute of Science and Technology (JAIST). He then served as an Assistant Professor at the Graduate School of Engineering, Osaka University, from 2016 to 2017. Since 2017, he has been with the University of Aizu, where he is currently a Senior Associate Professor. Prof. Su's research interests span trustworthy AI and cyber-physical security, including privacy-enhanced federated learning, AIoT/WiFi sensing, blockchain-assisted authentication and reputation, physical-layer key generation, and intrusion detection for critical infrastructures. He has published over 200 papers in international journals and conferences.

Keynote Speakers and Talks

Cong Zuo

Tenure-track Professor and Doctoral Supervisor
School of Cyberspace Security, Beijing Institute of Technology

Saturday 14 November 2026 | 09:30–10:10 (UTC+8)



Dynamic Symmetric Searchable Encryption with Conjunctive Queries and Malicious Server Traceability

Abstract

Dynamic symmetric searchable encryption (DSSE) enables efficient searches over encrypted data, but typically leaks search patterns, allowing adversaries to infer sensitive information. Approaches based on oblivious random-access memory (ORAM) can mitigate this leakage, but their high overhead hinders large-scale deployment. Dory uses a multi-server architecture and distributed point functions (DPFs) to reduce leakage and improve performance. However, it lacks a mechanism for identifying individual malicious servers: its adversarial model is ambiguous, and the protocol provides no way to determine which server has misbehaved. Dory also supports only single-keyword queries in a single-client setting, limiting its practicality. To address these limitations, we propose a more practical DSSE scheme that allows multiple clients to retrieve data using conjunctive queries. The scheme can trace individual malicious servers while hiding search patterns and providing forward and backward privacy. To achieve these goals, we introduce a new primitive, matrix-based multi-point retrieval (MMPR), to hide search patterns, together with a novel data structure, the malicious-resistant Bloom filter (MDBF). This data structure supports conjunctive queries and verifies the integrity of server responses, further strengthening security.

Biography

Cong Zuo is a tenure-track professor and doctoral supervisor at the School of Cyberspace Security, Beijing Institute of Technology. His research focuses on symmetric searchable encryption, data privacy protection, and applied cryptography. His work has appeared in leading information security journals and conferences, including TIFS, TDSC, CCS, and ESORICS. He has served as a session chair or program committee member for PROVSEC 2019, NSS 2019, PST 2021, PROVSEC 2021, ADIoT 2022, ACNS-SCI 2023, PST 2023, PST 2025, ICICS 2025, AIBlock 2026, SciSec 2026, and ICICS 2026. He has also served as Assistant Guest Editor for the MDPI Cryptography special issue "Emerging Cryptographic Protocols for Blockchain and Its Applications," and as a reviewer for leading venues including TIFS, TDSC, TSC, Eurocrypt, and Asiacrypt. In 2022, he was selected for a national program for outstanding young talent.

Keynote Speakers and Talks

Weizheng Wang

Professor, School of Cyberspace Security, Guangzhou University

Saturday 14 November 2026 | 10:40–11:20 (UTC+8)



A Safety Harness for Embodied Intelligence

Abstract

Embodied agents driven by foundation models are moving from text interaction into the physical world. Their safety therefore depends both on appropriate outputs and on keeping real-world actions under control. Signatures and authentication can establish a trusted command source, but cannot guarantee that conditions at execution time still match those under which authorization was granted. This talk presents a runtime safety harness framework for embodied intelligence developed by Academician Bingxing Fang's team. It first introduces the basic problem of runtime safety for embodied agents. A placement task in which authorization remains valid but the target container has been replaced illustrates the gap between trust in the command source and trust in the current situation. Drawing on the speaker's research in device authentication and cross-domain authentication, the talk then explains how an independent supervisory robot actively gathers evidence before critical actions, binds tasks, physical entities, and actions into verifiable evidence, and works with a protected module on the execution side to implement risk-triggered interruption and controlled recovery before an action begins. Finally, using an experimental platform comprising a quadruped execution robot and a wheeled supervisory robot, the talk discusses the approach's applicability relative to fixed-camera solutions and the challenges of practical deployment.

Biography

Weizheng Wang is a professor at the School of Cyberspace Security, Guangzhou University. He received his Ph.D. from City University of Hong Kong and conducted postdoctoral research at The Hong Kong Polytechnic University. He has led multiple projects funded by the National Natural Science Foundation of China. His research interests include cross-technology communication, wireless network and Internet of Things security, applied cryptography, and trusted authentication. He has published more than 20 papers in CCF A-ranked journals and conferences.

Keynote Speakers and Talks

Guolei Sun

Professor and Doctoral Supervisor
College of Computer Science, Nankai University

Saturday 14 November 2026 | 11:20–12:00 (UTC+8)

Compression Techniques for Large Multimodal Models

Abstract

Large multimodal models have grown in parameter count in recent years, demanding ever greater computational resources. Model compression aims to accelerate inference, reduce the memory required to run models, and lower their power consumption. This talk introduces two model compression techniques: ultra-low-bit quantization and pruning with extremely low parameter retention rates. Related papers have been published in or submitted to CCF A-ranked conferences, including NeurIPS and AAAI.

Biography

Guolei Sun received his Ph.D. from ETH Zurich and is a professor and doctoral supervisor at the College of Computer Science, Nankai University. He has been selected for a national young talent program. His research interests include large multimodal models, video understanding, model compression, and 3D vision. He has published more than 60 papers in SCI Q1 journals and CCF A-ranked venues, including multiple TPAMI and IJCV papers and several oral and spotlight presentations. He holds two invention or utility model patents. His papers have received more than 12,000 Google Scholar citations, with over 6,000 citations for his most-cited paper. His research has received support from the National Natural Science Foundation of China, Toyota, and OPPO. His awards include best paper awards at CVPR and ICCV workshops, first place in a CVPR workshop challenge on weakly supervised segmentation, and the Special Award of the Chinese Government Award for Outstanding Self-Financed Students Abroad. He was a core member of a Swiss national project on personalized medicine and related technologies. His research has been adopted by LPIXEL in Japan and featured in the UK science magazine New Scientist. He has served as an area chair or editorial board member for leading international AI conferences and journals, including ICLR and NeurIPS.